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SUB CODE: DSA0502

SUB NAME: QUERY PROCESSING FOR DATA SCIENCE

CO3 ASSESSMENT TOOL 1

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Question 1: E-Commerce Customer Transactions Dataset

An online retail company has collected customer transaction data from multiple sales channels. The management wants to identify the top-performing customer segments and products. However, the dataset contains several quality issues.

Data Issues:

- Missing Customer IDs
- Duplicate transactions
- Negative quantities
- Unit prices recorded in different currencies
- Age values such as 0, 150, and blanks
- Inconsistent city names (e.g., Chennai, chennai, CHENNAI)
- Incorrect purchase dates

Tasks:

1. Identify and classify all data quality issues.
2. Design a cleaning strategy for each issue.
3. Perform data standardization and transformation.
4. Analyze how cleaning affects total revenue calculations.
5. Generate a report discussing business risks if the data were not cleaned.

AIM: The aim of this project is to identify and clean data quality issues in an E-Commerce Customer Transactions dataset. The dataset contains missing values, duplicate records, invalid ages, negative quantities, inconsistent city names, multiple currencies, and incorrect purchase dates. Python and Pandas were used to clean, standardize, and transform the data to improve its quality for accurate business analysis.

OBJECTIVE:

- To identify different types of data quality issues.
- To clean inconsistent and invalid data.
- To standardize the dataset into a uniform format.
- To analyze how cleaning affects total revenue.
- To understand the business risks caused by poor-quality data.

DATASET DESCRIPTION:

The dataset contains customer transaction details collected from an online retail company. The attributes include Customer ID, Customer Name, Gender, Age, Product Category, Quantity, Unit Price, Currency, Purchase Date, and City. The dataset contains several inconsistencies such as missing values, duplicate records, invalid ages, incorrect dates, multiple currencies, and inconsistent city names, which must be cleaned before performing analysis.

TASK 1 – IDENTIFY AND CLASSIFY DATA QUALITY ISSUES

The following data quality issues were identified in the dataset.

Missing Customer IDs	- Some customer IDs are blank or missing.
Duplicate Transactions	- One or more customer transactions are repeated.
Negative Quantity	- Quantity values are less than zero.
Invalid Age	- Age values such as 0 and 150 are invalid.
Multiple Currencies	- Unit prices are recorded in INR, USD, and EUR.
InconsistentCity Names	- City names appear as Chennai, chennai, and CHENNAI.
IncorrectPurchaseDate	- Some dates are in an invalid format.

Observation

The dataset contains several data quality problems that affect the accuracy of data analysis and business decision-making. These issues must be corrected before calculating revenue or generating reports.

TASK 2 – CLEANING STRATEGY

The following cleaning techniques were applied to improve the quality of the dataset.

Issue	Cleaning Method
Missing Customer IDs	- Replaced missing values with "Unknown".
Duplicate Records	- Removed duplicate rows using drop_duplicates().
Invalid Age	- Removed records where age is 0 or greater than 120.
Negative Quantity	- Removed records having quantity less than zero.
Incorrect Purchase Date	- Converted dates into a standard date format.
Inconsistent City Names	- Converted all city names into Proper Case.
Multiple Currencies	- Converted all prices into INR.

Observation

Applying these cleaning methods improved the consistency and reliability of the dataset. Invalid and duplicate records were removed, making the data suitable for analysis.

TASK 3 – DATA STANDARDIZATION AND TRANSFORMATION

Standardization Performed

- City names were converted into Proper Case.
- Purchase dates were converted into a standard date format.
- Unit prices were converted into Indian Rupees (INR).
- Invalid records were removed.
- A new column named **Price_INR** was created.

Transformation Summary

Transformation	Result
City Standardization	- All city names follow the same format.
Currency Conversion	- All prices are converted into INR.
Date Conversion	- Purchase dates follow one standard format.
Invalid Data Removal	- Only valid customer records remain.

Observation

After transformation, the dataset became more consistent and easier to analyze. Standardization ensures accurate comparisons and reporting.

TASK 4 – EFFECT OF CLEANING ON TOTAL REVENUE

Formula : $\text{Revenue} = \text{Quantity} \times \text{Price}$

Revenue Comparison

Before Cleaning	After Cleaning
Revenue included duplicate transactions.	- Duplicate transactions removed.
Negative quantities affected calculations.	- Negative quantities removed.
Multiple currencies caused incorrect totals.	- All prices converted into INR.
Revenue was inaccurate.	- Accurate and reliable.

Analysis

Before cleaning, the revenue calculation was incorrect because duplicate records, negative quantities, and multiple currencies affected the total. After cleaning and converting all prices into INR, the revenue calculation became accurate and suitable for business reporting.

TASK 5 – BUSINESS RISKS IF DATA IS NOT CLEANED

Poor-quality data can create several business problems.

Business Risks

- Incorrect revenue calculations.
- Wrong customer segmentation.
- Duplicate transactions increase sales values.
- Mixed currencies produce inaccurate financial reports.
- Invalid ages reduce customer analysis accuracy.
- Incorrect dates affect monthly and yearly sales reports.
- Inconsistent city names produce incorrect regional analysis.
- Poor-quality data leads to wrong business decisions.

Discussion

Data cleaning is an important process before performing any business analysis. Clean data improves accuracy, consistency, and reliability. It helps organizations make better decisions, improve customer satisfaction, and generate correct financial reports.

RESULTS

- Missing values handled.
- Duplicate transactions removed.
- Invalid ages removed.
- Negative quantities removed.
- City names standardized.

- Dates standardized.
- Currency converted into INR.
- Revenue recalculated.

OUTPUT:

The screenshot shows a Google Colab notebook with the following code and output:

```
import pandas as pd
df = pd.read_excel("/content/co2.xlsx")
df.head()
```

	CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	Currency	PurchaseDate	City
0	C101	Raj Kumar	Male	28	Electronics	2	15000	INR	2025-01-10 00:00:00	Chennai
1	C102	Priya S	Female	35	Fashion	3	1200	INR	2025-01-11 00:00:00	chennai
2	C103	Arun V	Male	150	Electronics	1	300	USD	2025-01-12 00:00:00	CHENNAI
3	C104	Kavya R	Female	0	Grocery	-2	800	INR	2025-01-15 00:00:00	Coimbatore
4	C105	Rahul P	Male	42	Fashion	5	2500	INR	2025-01-18 00:00:00	Madurai

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
!pip install openpyxl
Requirement already satisfied: openpyxl in /usr/local/lib/python3.12/dist-packages (3.1.5)
Requirement already satisfied: et-xmlfile in /usr/local/lib/python3.12/dist-packages (from openpyxl) (2.0.0)
```

```
# Check missing values
print("==== Missing Values =====")
```

The screenshot shows the following code and output for data cleaning:

```
# ===== Missing Values =====
df.isnull().sum()
# CustomerID 1
# CustomerName 0
# Gender 0
# Age 0
# ProductCategory 0
# Quantity 0
# UnitPrice 0
# Currency 0
# PurchaseDate 0
# City: int64 0
```

```
# Check duplicate rows
print("==== Duplicate Rows =====")
print(df.duplicated().sum())
# 1
# Duplicate Rows =====
```

```
# Find negative quantities
print("==== Negative Quantity =====")
print(df[df["Quantity"] < 0])
# CustomerID CustomerName Gender Age ProductCategory Quantity UnitPrice \
# C104 Kavya R Female 0 Grocery -2 800
# Currency INR PurchaseDate 2025-01-15 00:00:00 City Coimbatore
```

The screenshot shows the final data cleaning steps and revenue calculation:

```
clean_df.to_csv("cleaned_customer_transactions.csv", index=False)
print("Cleaned dataset saved successfully!")
Cleaned dataset saved successfully!
```

```
# Revenue Before Cleaning
df["Revenue"] = df["Quantity"] * df["UnitPrice"]
print("Total Revenue Before Cleaning:")
print(df["Revenue"].sum())
# Total Revenue Before Cleaning:
# 92000
```

```
# Revenue After Cleaning
clean_df["Revenue"] = clean_df["Quantity"] * clean_df["UnitPrice"]
print("Total Revenue After Cleaning:")
print(clean_df["Revenue"].sum())
# Total Revenue After Cleaning:
# 463000
```

```
# Revenue Comparison
print("==== Revenue Comparison =====")
print("Before Cleaning :", df["Revenue"].sum())
print("After cleaning :", clean_df["Revenue"].sum())
# Revenue Comparison
```

CONCLUSION

This project demonstrated the importance of data cleaning in real-world business datasets. Various data quality issues such as missing values, duplicate records, invalid ages, inconsistent city names, incorrect dates, and multiple currencies were identified and corrected. After cleaning, the dataset became consistent, reliable, and suitable for analysis. Accurate and clean data helps organizations make informed business decisions, calculate revenue correctly, and improve overall data quality.

Question 2: Hospital Patient Records Dataset

A hospital wants to analyze patient admissions and treatment effectiveness. Data was entered manually by different departments, leading to inconsistencies.

Data Issues:

- Missing patient IDs
- Duplicate patient records
- Discharge dates before admission dates
- Null treatment costs
- Inconsistent diagnosis names
- Multiple formats of dates
- Unrealistic patient ages (>120)

Tasks:

1. Detects all inconsistencies and anomalies.
2. Develop rules for handling missing and invalid values.
3. Create a process to standardize diagnosis names.
4. Evaluate how cleaning changes average treatment cost and hospital stay duration.
5. Recommend data governance measures to prevent future error.

AIM: The aim of this project is to identify and clean inconsistencies in the Hospital Patient Records dataset by handling missing values, duplicate records, invalid ages, incorrect admission and discharge dates, inconsistent diagnosis names, and missing treatment costs using Python and Pandas.

OBJECTIVE:

- To identify data quality issues in hospital records.
- To clean missing and invalid values.
- To standardize diagnosis names.
- To analyze the effect of cleaning on treatment cost and hospital stay.
- To recommend data governance practices for maintaining high-quality hospital data.

DATASET DESCRIPTION

The Hospital Patient Records dataset contains information about patients admitted to a hospital. The dataset includes Patient ID, Name, Age, Gender, Admission Date, Discharge Date, Diagnosis, Treatment Cost, and Doctor ID. Since the data was entered manually by different departments, it contains several inconsistencies such as missing values, duplicate records, invalid ages, incorrect dates, and inconsistent diagnosis names. These issues were cleaned before analysis.

TASK 1 – DETECT ALL INCONSISTENCIES AND ANOMALIES

Data Quality Issue	Description
Missing Patient ID	- One patient record contains a missing Patient ID.
Duplicate Records	- One patient record is duplicated.

Invalid Age	- Age value greater than 120 is unrealistic.
Incorrect Dates	- One discharge date is earlier than the admission date.
Missing Treatment Cost	- Some treatment cost values are missing.
Inconsistent Diagnosis Names	- Diagnosis names appear as Diabetes, diabetes, Diabtes, and HYPERTENSION.

Observation

The dataset contained several inconsistencies that could affect hospital reports and analysis. These issues must be corrected before performing any calculations.

TASK 2 – DEVELOP RULES FOR HANDLING MISSING AND INVALID VALUES

Data Issue	Cleaning Rule
Missing Patient ID	- Replaced with "Unknown".
Duplicate Records	- Removed duplicate rows.

Missing Treatment Cost - Replaced missing values with the average treatment cost.

Invalid Age - Removed records with age greater than 120 years.

Incorrect Dates - Converted dates into a standard format and identified invalid records.

Observation

The cleaning process removed duplicate and invalid records while filling missing values. This improved the quality and consistency of the dataset.

TASK 3 – STANDARDIZE DIAGNOSIS NAMES

Standardization Performed

- "diabetes" was changed to "Diabetes".
- "Diabtes" was corrected to "Diabetes".
- "HYPERTENSION" was changed to "Hypertension".

Standardization Summary

Before Cleaning	After Cleaning
diabetes	Diabetes
Diabtes	Diabetes
HYPERTENSION	Hypertension

Observation

Standardizing diagnosis names ensures that all patients with the same disease are grouped together correctly during analysis and reporting.

TASK 4 – EFFECT OF CLEANING ON AVERAGE TREATMENT COST AND HOSPITAL STAY

The average treatment cost was calculated after handling missing values. The average hospital stay was calculated using the difference between the discharge date and admission date.

Analysis:

Before Cleaning

Missing treatment costs affected the average.

Incorrect dates affected hospital stay calculations.

Duplicate records affected analysis.

After Cleaning

Missing values were replaced, giving a reliable average.

Standardized dates provided accurate stay duration.

Duplicate records were removed.

Observation

After cleaning, the average treatment cost and hospital stay duration became more accurate and reliable. The cleaned dataset provides better support for hospital management and decision-making.

TASK 5 – DATA GOVERNANCE MEASURES

To prevent similar data quality issues in the future, the following data governance measures are recommended:

- Make Patient ID a mandatory field.
- Validate patient age during data entry.
- Ensure discharge dates cannot be earlier than admission dates.
- Use drop-down lists for diagnosis names to avoid spelling mistakes.
- Validate treatment cost to allow only numeric values.
- Perform regular data quality checks.
- Train hospital staff on proper data entry procedures.
- Maintain backup and audit records for all patient information.

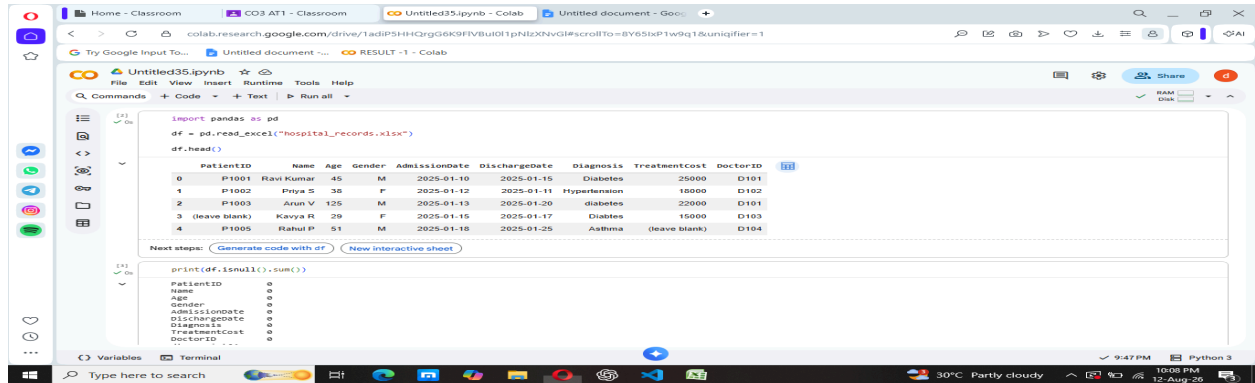
Discussion

Implementing these data governance measures will improve data accuracy, reduce manual errors, and ensure reliable patient records for analysis and reporting.

RESULTS

- Missing Patient IDs handled.
- Duplicate records removed.
- Invalid ages removed.
- Missing treatment costs filled.
- Diagnosis names standardized.
- Admission and discharge dates standardized.
- Average treatment cost calculated.
- Hospital stay duration calculated.

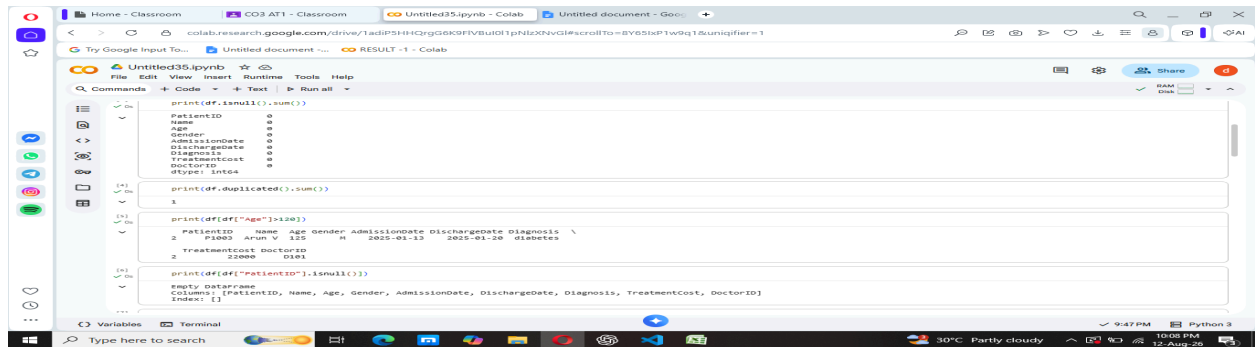
OUTPUT:



Colab notebook interface showing the initial data loading step. The code imports pandas and reads an Excel file named 'hospital_records.xlsx'. The output displays the first five rows of the DataFrame.

```
import pandas as pd
df = pd.read_excel("hospital_records.xlsx")
df.head()
```

	PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID
0	P1001	Ravi Kumar	45	M	2025-01-10	2025-01-15	Diabetes	25000	D101
1	P1002	Priya S	38	F	2025-01-12	2025-01-11	Hypertension	18000	D102
2	P1003	Arun V	125	M	2025-01-13	2025-01-20	diabetes	22000	D101
3	(leave blank)	Kavya R	28	F	2025-01-15	2025-01-17	Diabetes	15000	D103
4	P1005	Rahul P	51	M	2025-01-16	2025-01-25	Asthma	(leave blank)	D104



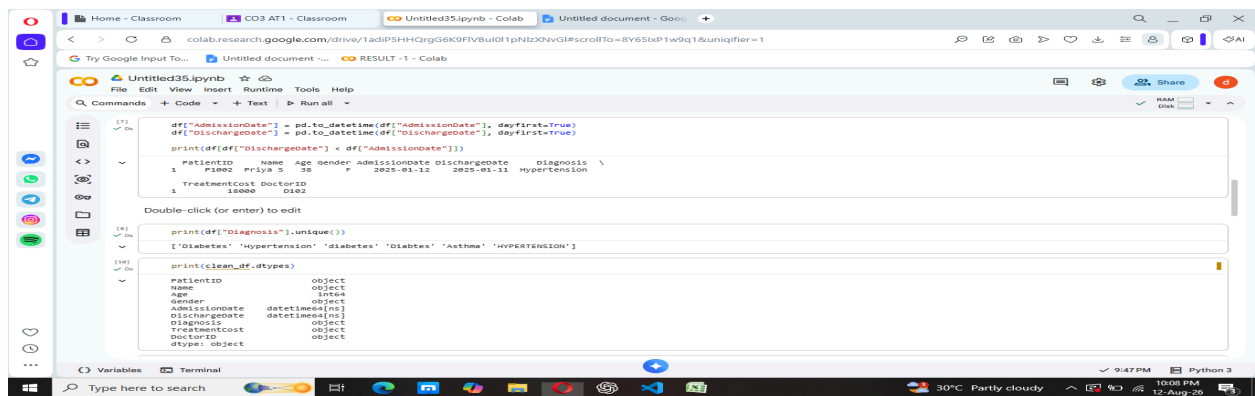
Colab notebook interface showing data cleaning steps. The code checks for null values, removes duplicates, and filters out rows with missing 'Age' values. It also checks for null values in the 'PatientID' column.

```
print(df.isnull().sum())

print(df.duplicated().sum())

print(df[df["Age"]>120])

print(df[df["PatientID"].isnull()])
```



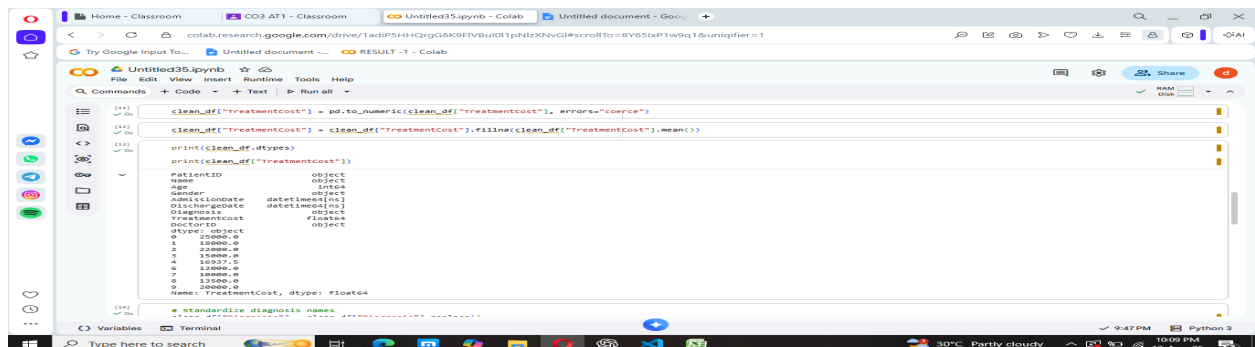
Colab notebook interface showing date conversion and unique diagnosis values. The code converts 'AdmissionDate' and 'DischargeDate' to datetime objects. It also prints the unique values for the 'Diagnosis' column and the data types of the DataFrame.

```
df["AdmissionDate"] = pd.to_datetime(df["AdmissionDate"], dayfirst=True)
df["DischargeDate"] = pd.to_datetime(df["DischargeDate"], dayfirst=True)

print(df[df["DischargeDate"] < df["AdmissionDate"]])

print(df["Diagnosis"].unique())

print(clean_df.dtypes)
```



Colab notebook interface showing data cleaning and standardization. The code converts 'TreatmentCost' to numeric, fills missing values with the mean, and standardizes diagnosis names.

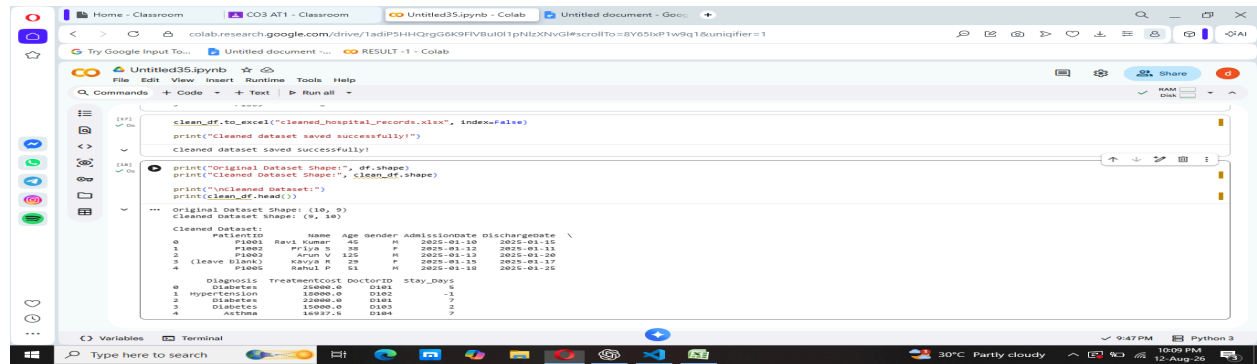
```
clean_df["TreatmentCost"] = pd.to_numeric(clean_df["TreatmentCost"], errors="coerce")

clean_df["TreatmentCost"] = clean_df["TreatmentCost"].fillna(clean_df["TreatmentCost"].mean())

print(clean_df.dtypes)

print(clean_df["TreatmentCost"])

standardize_diagnosis_names
```



```
clean_df.to_excel("cleaned_hospital_records.xlsx", index=False)
print("Cleaned dataset saved successfully!")
Cleaned dataset saved successfully!

print("Original Dataset Shape:", df.shape)
print("Cleaned Dataset Shape:", clean_df.shape)
print("\nCleaned Dataset:")
print(clean_df.head())

Original Dataset Shape: (10, 9)
Cleaned Dataset Shape: (9, 10)

Cleaned Dataset:
  PatientID  Name  Age  Gender  AdmissionDate  DischargeDate  \
0  P1001  Ravi Kumar  45  H  2023-01-10  2023-01-15
1  P1002  Priya S  30  F  2023-01-12  2023-01-18
2  P1003  Arun V  120  H  2023-01-13  2023-01-20
3  (leave blank)  Nanya S  25  F  2023-01-15  2023-01-17
4  P1005  Rahul P  51  H  2023-01-16  2023-01-22

  Diagnosis  TreatmentCost  DoctorID  Stay_Days
0  Diabetes  25000.0  D101  5
1  Hypertension  18000.0  D102  -1
2  Diabetes  22000.0  D103  7
3  Diabetes  15000.0  D104  2
4  Asthma  16000.0  D104  7
```

CONCLUSION

This project demonstrated the importance of data cleaning in healthcare datasets. Various data quality issues such as missing values, duplicate records, unrealistic ages, inconsistent diagnosis names, incorrect dates, and missing treatment costs were identified and corrected. After cleaning, the dataset became accurate and reliable for hospital reporting and analysis. Clean healthcare data helps improve patient management, supports informed decision-making, and enhances the overall quality of healthcare services.